William F. Li

william__li@hms.harvard.edu
williamfli.github.io
scholar.google.com/citations?user=avkQHcwAAAAJ

Education

Harvard Medical School — Boston, MA
MD — 2024 – present

Harvard University — Cambridge, MA
PhD in Biophysics — 2026 – present

Massachusetts Institute of Technology — Cambridge, MA
SB in Physics and Computer Science & Engineering — 2020 – 2024
GPA: 5.00/5.00, Phi Beta Kappa, Minor in Biology

C. Leon King High School — Tampa, FL
International Baccalaureate Diploma — 2016 – 2020
GPA: 4.00/4.00; IB Higher Level Courses in Mathematics, Physics, and English

Experience

Harvard Department of Chemistry and Chemical Biology
Graduate Researcher — 2024 – present

- Principal Investigator: Xiaowei Zhuang
- Research topic:
  - 3D-genome imaging of the human brain

MIT Computer Science and Artificial Intelligence Laboratory
Undergraduate Researcher — 2022 – 2024

- Principal Investigator: Manolis Kellis
- Research topic:
  - Genetics of Alzheimer's disease heterogeneity

MIT Research Laboratory of Electronics
Undergraduate Researcher — 2020 – 2022

- Principal Investigators: Marin Soljačić, Steven Johnson
- Research topics:
  - X-ray imaging with nanophotonic scintillators
  - Computational imaging with compressed sensing and end-to-end inverse design

Awards

- National Institute of Aging F30 MD-PhD Fellowship — 2026 – 2032
- Harvard Medical School Summer Research Fellowship — 2025
- Phi Beta Kappa Liberal Arts and Sciences Honor Society — 2024
- Sigma Pi Sigma Physics and Astronomy Honor Society — 2024
- MIT News profile, MIT homepage (www.mit.edu) spotlight for March 15, 2024 — 2024
- Gates Cambridge finalist (Chemistry, 3D-genome simulation), MIT nomination for Rhodes and Marshall — 2023
- Medical College Admission Test (MCAT) Perfect Score (528/528) — 2023
- MIT EECS SuperUROP Outstanding Research Award, awarded to 2 in cohort — 2023
- *Optics Express* Editors' Pick — 2023
- Eric and Wendy Schmidt Center funded Research and Innovation Scholar — 2022 – 2023
- MIT UROP funded research for 8 semesters and 3 summers — 2020 – 2024
- USA Astronomy and Astrophysics National Team, ranked 8th nationally — 2020
- US Physics Team, USA Physics Olympiad Gold, top 20 nationally — 2019
- 2-time US National Chemistry Olympiad National Exam Qualifier, 1-time Tampa Bay 1st place — 2019, 2020
- Sunshine Scholar (Florida top STEM students) — 2019

- 2-time National AP Scholar (5/5 on 19 of 19 AP exams) 2018, 2019
- USA Computing Olympiad Gold Division 2018
- 3-time AMC 12 Distinguished Honor Roll (top 1%), 1-time Florida 1st place 2018, 2019, 2020
- 2-time USA Math Olympiad Qualifier, 1-time Junior Math Olympiad Qualifier 2017, 2019, 2020
- National Mathcounts Qualifier, Florida 3rd place, Tampa Bay 1st place 2016

Publications

3. Liu Z, Zhang S, James BT, Galani K, Mangan RJ, Fass SB, Liang C, Wagle MM, Boix CA, Tanigawa Y, Yun S, Sung Y, Xiong X, Sun N, Hou L, Wohlwend M, Qiu M, Han X, Xiong L, Preka E, Huang L, **Li WF**, Ho LL, Grayson A, Mantero J, Kozlenkov A, Mathys H, Chen T, Dracheva S, Bennett DA, Tsai LH, Kellis M. Single-cell multiregion epigenomic rewiring in Alzheimer's disease progression and cognitive resilience. *Cell*. 2025;188(18):4980–5002. doi:10.1016/j.cell.2025.06.031

2. Arya G, **Li WF**, Roques-Carmes C, Soljačić M, Johnson SG, Lin Z. End-to-End Optimization of Metasurfaces for Imaging with Compressed Sensing. *ACS Photonics*. 2024;11(5):2077–2087. doi:10.1021/acsphotonics.4c00259

1. **Li WF**, Arya G, Roques-Carmes C, Lin Z, Johnson SG, Soljačić M. Transcending shift-invariance in the paraxial regime via end-to-end inverse design of freeform nanophotonics. *Optics Express.* 2023;31(15):24260–24272. **Editors' Pick**. doi:10.1364/OE.492553

Manuscripts in Progress

2. **Li WF**, Mohammed N, Bennett DA, Kellis M, Tanigawa Y. Dissecting Alzheimer's disease heterogeneity by cross-trait polygenic prediction. *bioRxiv.* Preprint posted online May 2026. doi:10.64898/2026.05.15.725551v1. In review.

1. Tian X, Fabiha T, **Li WF**, Dey KK, Kellis M, Tanigawa Y. PRISM: ancestry-aware integration of tissue-specific genomic annotations enhances the transferability of polygenic scores. *bioRxiv.* Preprint posted online November 2025. doi:10.1101/2025.11.13.688144. In review.

Patents

1. Soljačić M, Roques-Carmes C, Rivera N, Lin Z, **Li WF**, inventors; Massachusetts Institute of Technology, assignee. Nanophotonic Scintillators for High-Energy Particles Detection, Imaging, and Spectroscopy. U.S. Patent Application US20250137942A1. August 2022.

Presentations

- Harvard Medical School Soma Weiss Research Day; March 2026; Boston, MA.
- American Society of Human Genetics; October 2025; Boston, MA.
- Harvard/MIT MD-PhD Program Retreat; September 2025; Cape Cod, MA.
- Conference on Lasers and Electro-Optics; May 2023; San Jose, CA.
- Broad Institute Scientific Retreat; December 2022; Boston, MA.
- MIT EECS SuperUROP Showcase; December 2022; Cambridge, MA.
- MIT PRISM Undergraduate Physics Conference; September 2022; Cambridge, MA.
- Conference on Lasers and Electro-Optics; May 2022; San Jose, CA.

Service and Teaching

- MIT Epsilon Theta/Xi Fellowship graduate resident advisor
- Harvard College Quincy House non-resident premedical tutor
- Northeast Regional Vascular and Interventional Radiology Symposium moderator
- MIT Prehealth mock interviewer
- Junior reviewer for *Nature Communications*

- MIT Department of Physics: scribe, tutor, freshman pre-orientation program
- MIT Department of Electrical Engineering and Computer Science associate advisor
- Massachusetts General Hospital volunteer in patient transport and emergency department
- UPchieve volunteer tutor
- MIT Students for Open and Universal Learning course recruiter
- AwesomeMath Summer Program teaching assistant for combinatorics
- Byrd Alzheimer's Institute research volunteer with Prof. Laura Blair's lab
- Melodies for Life Assisted Living Music Group: volunteer cello, coordinator

Leadership and Activities

- **Medical/Graduate School:** Interventional Radiology Interest Group (mentorship & event chair), Radiology Interest Group, American Society of Human Genetics, American Medical Association, Massachusetts Medical Society, Run Club (2024 Newburyport Half Marathon, 2024 Cambridge Half Marathon, 2025 Multiple Myeloma Research Foundation 5k)
- **Undergraduate:** MIT Premedical Society (collegiate relations co-chair), Journal Club in Genomics (founder/organizer), MIT Next House Thanksgiving Committee, Institute of Electrical and Electronics Engineers
- **High School:** Florida Student Association of Mathematics (state co-president), Mu Alpha Theta (president), Science National Honor Society (president), Orchestra (all-county principal cello), Swim (varsity team)

Selected Coursework

- **Harvard**
  - **Clinical Medicine (Massachusetts General Hospital):** Obstetrics & Gynecology, Pediatrics, Psychiatry, Neurology, Radiology
  - **Preclinical Medicine:** Foundations (Histology, Anatomy, Cell Biology & Biochemistry, Genetics, Pharmacology, Immunology, Microbiology, Developmental Biology, Pathology, Cancer); Integrated Human Pathophysiology I–III (Hematology, Gastroenterology, Cardiology, Pulmonology, Endocrinology, Nephrology); Essentials of the Profession (Health Policy, Social Medicine, Clinical Epidemiology & Population Health, Medical Ethics & Professionalism); Immunity in Defense and Disease (Rheumatology, Dermatology, Infectious Disease, Allergy); Mind, Brain and Behavior (Neurology, Psychiatry)
- **MIT**

  *Advanced Standing Exam

  †Graduate Course
  - **Physics:** Multivariable Calculus*, Differential Equations*, Linear Algebra*, Electricity and Magnetism*, Vibrations and Waves*, Quantum Physics I*, Quantum Physics II, Classical Mechanics II, Statistical Physics I, Experimental Physics I, Signal Processing, Fundamentals of Statistics†, Mathematical Methods in Nanophotonics†
  - **Computer Science & Engineering:** Introduction to Computer Science and Programming in Python*, Introduction to Computational Thinking and Data Science, Mathematics for Computer Science, Introduction to Machine Learning, Fundamentals of Programming, Software Construction, Introduction to Algorithms, Design and Analysis of Algorithms, Computation Structures, Computer Systems Engineering
  - **Computational Biology:** Introductory Biology*, Principles of Chemical Science*, General Biochemistry, Genetics, Cell Biology, Organic Chemistry I, Organic Chemistry II, Thermodynamics I, Thermodynamics II and Kinetics, Thermodynamics of Biomolecular Systems, Fundamentals of Experimental Molecular Biology, Laboratory Chemistry, Cellular Neurobiology, Seminar in Undergraduate Advanced Research (SuperUROP), Computational Biology, Neurogenomics/Advanced Topics in Artificial Intelligence†, Topics in Computational Molecular Biology†